

# AERO40005 PROPERTIES OF MATERIALS

## Tutorial questions

By Dr. Q Li

Note: Questions 1-4 are for "Elasticity"

Questions 5-7 are for "Plasticity"

### Question 1

- What are the three most common crystal structures for metals? Name the densely packed planes and directions for the two cubic systems (for "Plasticity").
- Gold has an fcc structure and a lattice constant of 0.40788 nm. How many atoms are in the fcc unit cell? Calculate the value for the atomic radius of gold based on this information. Indicate the following planes and direction (100), (111), [110], [210] in the fcc lattice.
- Pure iron is polymorphic and changes from bcc to fcc at 912°C. Calculate the density change associated with this structural change. Lattice parameter of  $\alpha$ -iron (bcc) is  $a=0.293$  and of  $\gamma$ -iron (fcc)  $a=0.363$  nm.

(a) HCP (Hexagonal closely Packed crystals)

FCC (Face Central Cubic)

- densely packed planes {111}
- densely packed directions  $\langle 110 \rangle$

BCC (Body Central Cubic)

- no densely packed planes
- densely packed directions  $\langle 111 \rangle$

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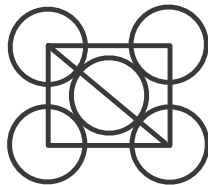
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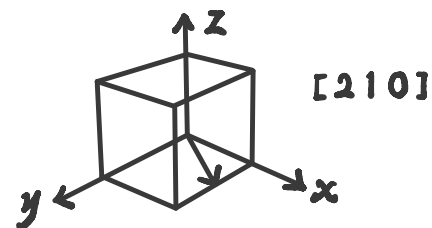
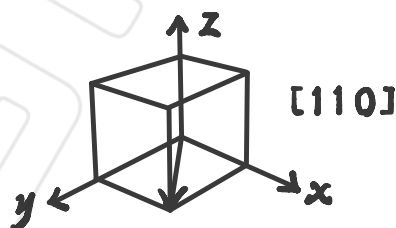
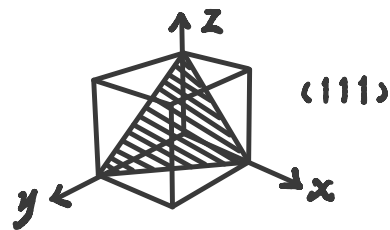
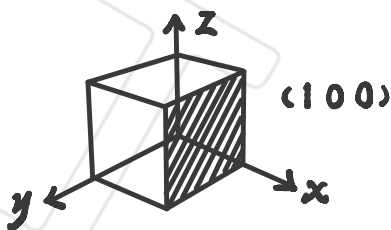
(b) 
$$N = \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$$



According to the diagram,

$$a^2 + a^2 = [(1+2+1)r]^2 = 16r^2$$

$$\therefore r = \frac{\sqrt{2}}{4}a = 0.14421 \text{ nm}$$



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$$(c) \quad N_{bcc} = \frac{1}{8} \times 8 + 1 = 2$$

$$N_{fcc} = \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$$

$$\rho_{bcc} = \frac{m}{V} = \frac{2M}{(0.293 \text{ nm})^3}$$

$$\rho_{fcc} = \frac{m}{V} = \frac{4M}{(0.363 \text{ nm})^3}$$

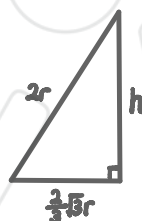
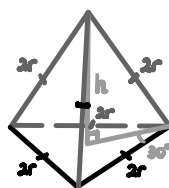
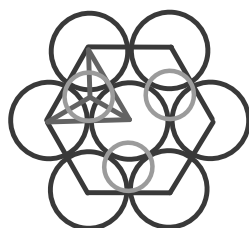
$$\therefore \Delta \rho = \frac{\rho_{fcc} - \rho_{bcc}}{\rho_{bcc}} \times 100\% = 5.175\%$$

## Question 2

Magnesium has a hexagonal close-packed structure.

- (a) Assuming that the atoms can be represented as hard spheres, calculate the percentage of the available volume occupied by atoms in each material, and comment on your answer.
- (b) The density of magnesium crystal is  $1740 \text{ kg m}^{-3}$  respectively. Calculate (i) the dimensions of the unit cells, and hence (ii) the radii of magnesium atoms.
- Hint: For magnesium of HCP structure, 1 atom weighs  $24.31 \text{ amu}$  (this information can be taken from the periodic table,  $1 \text{ amu} = 1.661 \times 10^{-27} \text{ kg}$ ).

(a)



$$\therefore h = \frac{2}{3}\sqrt{6}r$$

$$N = 3 + 12 \times \frac{1}{6} + 2 \times \frac{1}{2} = 6$$

$$V = A(2h) = 6 \left( \frac{1}{2} \times 2r \times \sqrt{3}r \right) \times \left( 2 \times \frac{2}{3}\sqrt{6}r \right) = 24\sqrt{2}r^3$$

$$V_{\text{atom}} = 6 \left( \frac{4}{3}\pi r^3 \right) = 8\pi r^3$$

$$\therefore \%AV = \frac{V_{\text{atom}}}{V} \times 100\% = 74.0\%$$

b) i)  $V = \frac{m}{\rho} = \frac{6 \times 24.31 \times 1.661 \times 10^{-27} \text{ kg}}{1740 \text{ kg/m}^3} = 1.3924 \times 10^{-28} \text{ m}^3$

ii)  $V = 24\sqrt{2}r^3 \quad \therefore r = \sqrt[3]{\frac{V}{24\sqrt{2}}} = 1.6008 \times 10^{-10} \text{ m}$

$$\therefore h = \frac{2}{3}\sqrt{6}r = 2.6141 \times 10^{-10} \text{ m}$$

### Question 3

The potential energy of a pair of adjacent atoms in a solid is

$$U = -\frac{A}{r^m} + \frac{B}{r^n}$$

where  $r$  is the distance between atom centres.

- (a) Express this relationship in terms of the equilibrium spacing,  $r_0$ , of the atoms and the potential energy at this spacing,  $U_0$ .
- (b) For  $n=6$  and  $m=2$  and a bond energy of  $-3.2$  eV and  $r_0 = 0.4$  nm calculate  $A$  and  $B$  (N.B.  $1\text{eV} = 1.602 \times 10^{-19}$  Joules).

$$(a) U_0 = -\frac{A}{(r_0)^m} + \frac{B}{(r_0)^n}$$

$$(b) \frac{dU}{dr} = mAr^{-m-1} - nBr^{-n-1} = 0 \text{ when } U = U_0$$

$$\therefore 2A(0.4 \times 10^{-9}\text{m})^{-3} - 6B(0.4 \times 10^{-9}\text{m})^{-7} = 0 \quad \therefore B = \frac{1}{3}A(0.4 \times 10^{-9}\text{m})^4$$

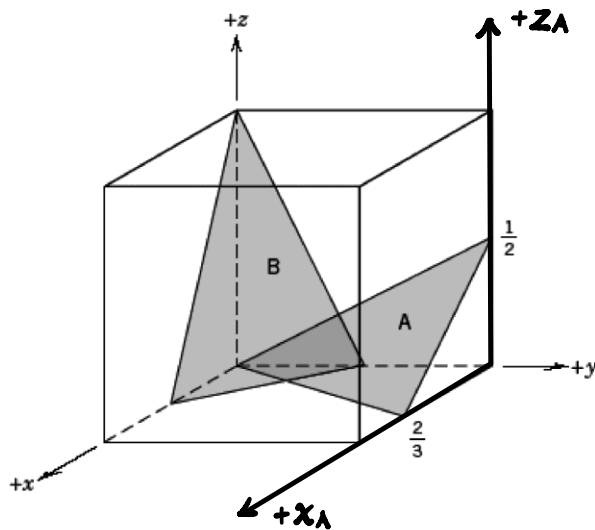
$$\therefore -3.2 \times 1.602 \times 10^{-19}\text{J} = -\frac{A}{(0.4 \times 10^{-9}\text{m})^2} + \frac{B}{(0.4 \times 10^{-9}\text{m})^6} = -\frac{2}{3}\frac{A}{(0.4 \times 10^{-9}\text{m})^2}$$

$$\therefore A = 1.230 \times 10^{-37}$$

$$B = 1.050 \times 10^{-75}$$

#### Questions 4

Determine the Miller indices for the planes shown in the following unit cell:



For A,

intercept  $\left[ \frac{2}{3} \quad \bar{1} \quad \frac{1}{2} \right]$

reciprocals  $\left( \frac{3}{2} \quad \bar{1} \quad 2 \right)$

$\therefore$  plane  $(3 \quad \bar{2} \quad 4)$

For B,

intercept  $\left[ \frac{1}{2} \quad \frac{1}{2} \quad 1 \right]$

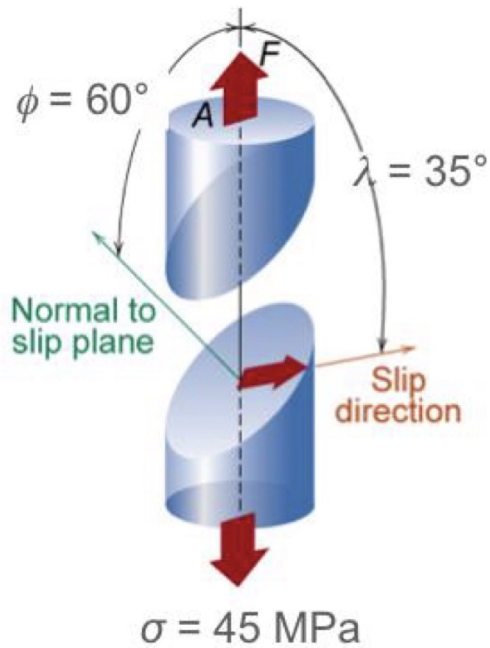
reciprocals  $(2 \quad 2 \quad 1)$

$\therefore$  plane  $(2 \quad 2 \quad 1)$

### Question 5

A bar of cross-section A is made from a single-crystal metallic material, with one active slip system identified as shown in the figure below. Critical resolved shear stress  $\tau_{crss} = 20.7 \text{ MPa}$

- (a) Will this single crystal yield?
- (b) If not, what stress is needed?



$$\begin{aligned} \text{(a)} \quad \tau &= \frac{F_s}{A_s} \\ &= \frac{F \cos \lambda}{\frac{A}{\cos \phi}} \\ &= \frac{F}{A} \cos \lambda \cos \phi \\ &= 6 \cos \lambda \cos \phi \\ &= 18.43 \text{ MPa} \\ &< \tau_{crss} \end{aligned}$$

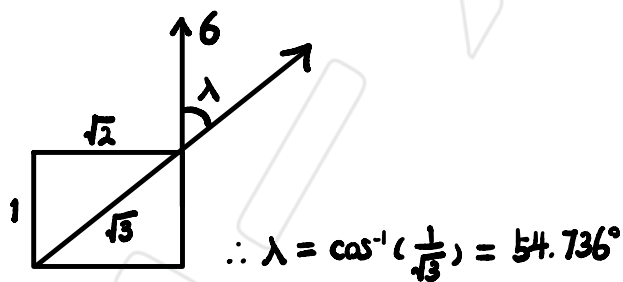
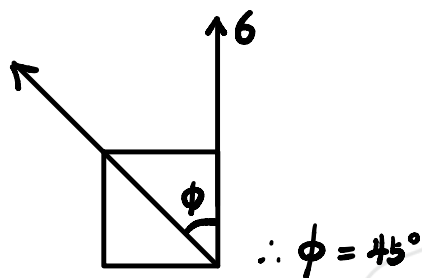
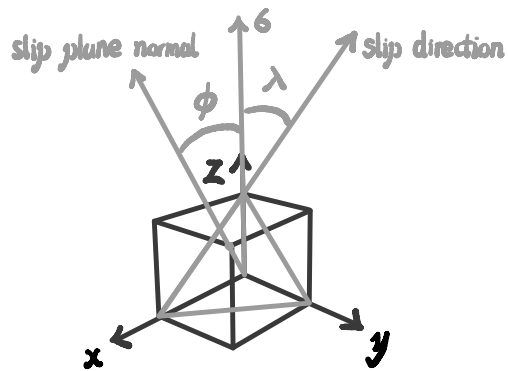
$$\text{(b)} \quad 6 \cos \lambda \cos \phi > \tau_{crss}$$

$$6 \cos \lambda \cos \phi > \frac{\tau_{crss}}{\cos \lambda \cos \phi}$$

$$\therefore 6 \cos \lambda \cos \phi > 50.54 \text{ MPa}$$

### Question 6

Consider a single crystal of silver oriented such that a tensile stress is applied along a [001] direction. If slip occurs on a (111) plane and in a  $[\bar{1}01]$  direction, and is initiated at an applied tensile stress of 1.4 MPa, calculate the critical resolved shear stress.



$$\therefore \tau = 6 \cos \lambda \cos \phi = 0.5715 \text{ MPa}$$



### Question 7

- (a) Describe in your own words the four strengthening mechanisms.
- (b) The lower yield point for iron with an average grain diameter of  $5 \times 10^{-2}$  mm is 135 MPa. At a grain diameter of  $8 \times 10^{-3}$  mm, the yield point increases to 260 MPa. At what grain diameter will the yield point be 205 MPa?

(a) ...

$$(b) \sigma_y = a + kd^{-\frac{1}{2}}$$

$$\therefore 135 \times 10^6 \text{ Pa} = a + k(5 \times 10^{-5} \text{ m})^{-\frac{1}{2}}$$

$$260 \times 10^6 \text{ Pa} = a + k(8 \times 10^{-6} \text{ m})^{-\frac{1}{2}}$$

$$\therefore a = 51666666.6$$

$$k = 589255.651$$

$$\therefore \text{When } \sigma_y = 205 \text{ MPa}$$

$$d = 1.4768431 \times 10^{-5} \text{ m}$$

$$= 1.48 \times 10^{-2} \text{ mm}$$